Derivations and numerical checks for optimal focusing

Companion notes to Secs. 3.1–3.2 of "Fundamental limits in photonics and electromagnetics: a tutorial"

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Abstract

These notes collect the derivations behind the free-space focusing results (tutorial Sec. 3.1) and the complex-media focusing results (Sec. 3.2) of the tutorial by Miller and Monticone, at a level of detail that would be out of place in a tutorial of that breadth. Every closed-form result is checked numerically by a script in the accompanying repository. Each check is independent of the derivation it tests, e.g. brute-force quadrature with explicit polarization vectors, or a full multiple-scattering model. The achieved agreement is quoted in green alongside each result. Several results here are new relative to the tutorial text: closed forms for the full (anisotropic) focal-plane fields, the exact energy penalties for super-oscillatory spots, a closed form for the three-dimensional space-time focus, and explicit symmetry conditions for the point-dipole media used to test Sec. 3.2.

Contents

1 Conventions **2**

2 Focusing from a cone of plane waves (tutorial Sec. 3.1.2) **2**
2.1 Plane-wave expansion and power . . . 2
2.2 Optimal coefficients and the focal matrix . . . 3
2.3 Focal-point intensity maxima . . . 4

3 The optimal fields across the focal plane **5**
3.1 General numerical aperture . . . 5
3.2 Closed forms at 2π solid angle . . . 6

4 Full 4π solid angle (tutorial Sec. 3.1.3) **8**
4.1 Herglotz form and the imaginary part of the Green's function . . . 8
4.2 Vector spherical waves . . . 8

5 Super-oscillatory spots (tutorial Sec. 3.1.4) **9**
5.1 Minimum energy under point constraints . . . 9
5.2 Zero ring in two dimensions . . . 10

6 Space-time focusing (tutorial Sec. 3.1.5) **10**

7 Focusing in complex media (tutorial Sec. 3.2) 11
7.1 The bound and the optimal currents . . . 11
7.2 A point-dipole medium with controllable symmetries . . . 12
7.3 Numerical results . . . 12
7.4 Partial time reversal on a closed surface . . . 13
7.5 Space-time version . . . 13

8 Index of numerical checks 14

1 Conventions

Fields are complex amplitudes with time dependence $e^{-i\omega t}$, the time-averaged Poynting vector is $\frac{1}{2}\,\mathrm{Re}(\mathbf{E}\times\mathbf{H}^*)$, and $k=\omega/c$. The electric energy density at a point is $U=\frac{1}{4}\varepsilon_0|\mathbf{E}|^2$, as in the tutorial. The numerical codes use natural units $\varepsilon_0=\mu_0=c=Z_0=1$ and report dimensionless ratios such as $(2\pi c/k^2)\,\varepsilon_0|\mathbf{E}(0)|^2/P$.

Spherical Bessel functions are denoted j_ℓ, cylindrical ones J_n, and $\mathrm{Si}(u)=\int_0^u \sin t/t\,\mathrm{d}t$. Two integral identities are used repeatedly:

$$\int_{-\pi}^{\pi} e^{iu\cos\varphi}\cos(n\varphi)\,\mathrm{d}\varphi = 2\pi i^n J_n(u), \qquad \int_{-\pi}^{\pi} e^{iu\cos\varphi}\sin(n\varphi)\,\mathrm{d}\varphi = 0, \tag{1}$$

$$\int_0^1 x^{\nu+1}(1-x^2)^{\mu} J_\nu(ax)\,\mathrm{d}x = 2^\mu\Gamma(\mu+1)\,a^{-\mu-1}J_{\nu+\mu+1}(a) \qquad \text{(Sonine)}, \tag{2}$$

the first being the Jacobi–Anger ("Hansen–Bessel") identity, and the second Sonine's first finite integral ($\mathrm{Re}\,\nu>-1$, $\mathrm{Re}\,\mu>-1$). With $j_\ell(a)=\sqrt{\pi/(2a)}\,J_{\ell+1/2}(a)$, the special cases needed below are

$$\begin{aligned}
\int_0^1 \frac{\beta J_0(a\beta)}{\sqrt{1-\beta^2}}\mathrm{d}\beta &= j_0(a), & \int_0^1 \beta\sqrt{1-\beta^2}\,J_0(a\beta)\mathrm{d}\beta &= \frac{j_1(a)}{a},\\
\int_0^1 \frac{\beta^3 J_2(a\beta)}{\sqrt{1-\beta^2}}\mathrm{d}\beta &= j_2(a), & \int_0^1 \beta^2\sqrt{1-\beta^2}\,J_1(a\beta)\mathrm{d}\beta &= \frac{j_2(a)}{a},
\end{aligned} \tag{3}$$

together with the elementary results $\int_0^s \beta J_0(a\beta)\mathrm{d}\beta = sJ_1(as)/a$, $\int_0^s \beta^2 J_1(a\beta)\mathrm{d}\beta = s^2J_2(as)/a$, and $\int_0^s \beta^3 J_2(a\beta)\mathrm{d}\beta = s^3J_3(as)/a$ (all from $\frac{\mathrm{d}}{\mathrm{d}x}[x^nJ_n(x)]=x^nJ_{n-1}(x)$), and $\int_0^s \beta^3 J_0(a\beta)\mathrm{d}\beta = s^3J_1(as)/a - 2s^2J_2(as)/a^2$.

The extremal-eigenvalue lemma. Every bound in these notes is an instance of the following. For a linear functional $f(c)=\langle v,c\rangle$ and a positive weight g,

$$\frac{|\langle v,c\rangle|^2}{\langle c, g\,c\rangle} \le \langle v, g^{-1}v\rangle, \qquad \text{with equality iff } c\propto g^{-1}v, \tag{4}$$

by the Cauchy–Schwarz inequality applied to $g^{1/2}c$ and $g^{-1/2}v$. Equivalently, the quadratic form $c^\dagger(vv^\dagger)c$ is rank one, so its Rayleigh quotient is maximized by its only nonzero eigenvector.

2 Focusing from a cone of plane waves (tutorial Sec. 3.1.2)

2.1 Plane-wave expansion and power

A field produced by sources in $z<0$ and observed in a region free of sources is a superposition of propagating plane waves (evanescent components are excluded, as they are absent at far-field

distances from the sources):

$$\mathbf{E}(\mathbf{x}) = \sum_p \int \mathrm{d}^2\mathbf{k}_\parallel \, c_p(\mathbf{k}_\parallel)\, \hat{\mathbf{e}}_p(\mathbf{k}_\parallel)\, e^{i\mathbf{k}\cdot\mathbf{x}}, \qquad \mathbf{H}(\mathbf{x}) = \frac{1}{Z_0} \sum_p \int \mathrm{d}^2\mathbf{k}_\parallel \, c_p(\mathbf{k}_\parallel)\, \hat{\mathbf{k}} \times \hat{\mathbf{e}}_p(\mathbf{k}_\parallel)\, e^{i\mathbf{k}\cdot\mathbf{x}}, \tag{5}$$

with $\mathbf{k} = \mathbf{k}_\parallel + k_z\hat{\mathbf{z}}$, $k_z = \sqrt{k^2 - |\mathbf{k}_\parallel|^2} \geq 0$, and $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{k}})$ a right-handed orthonormal triad. Numerical aperture $\sin\theta$ restricts $|\mathbf{k}_\parallel| \leq k \sin\theta$. The z-directed power through a plane is

$$\begin{aligned} P &= \frac{1}{2}\,\mathrm{Re}\int (\mathbf{E}\times\mathbf{H}^*)\cdot\hat{\mathbf{z}}\,\mathrm{d}x\,\mathrm{d}y = \frac{(2\pi)^2}{2Z_0}\,\mathrm{Re}\sum_{p,p'}\int \mathrm{d}^2\mathbf{k}_\parallel\, c_p c_{p'}^*\,[\hat{\mathbf{e}}_p \times (\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{p'}^*)]\cdot\hat{\mathbf{z}} \\ &= \frac{2\pi^2}{Z_0}\sum_p \int \mathrm{d}^2\mathbf{k}_\parallel\, \frac{k_z}{k}|c_p(\mathbf{k}_\parallel)|^2, \end{aligned} \tag{6}$$

where the spatial integral produced $(2\pi)^2\delta(\mathbf{k}_\parallel - \mathbf{k}'_\parallel)$ and $\hat{\mathbf{e}}_p\times(\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{p'}^*) = \hat{\mathbf{k}}(\hat{\mathbf{e}}_p\cdot\hat{\mathbf{e}}_{p'}^*) - \hat{\mathbf{e}}_{p'}^*(\hat{\mathbf{e}}_p\cdot\hat{\mathbf{k}}) = \hat{\mathbf{k}}\,\delta_{pp'}$. [numerical check: `power_normalization_check.py`: direct Poynting integration of FFT-synthesized fields with random complex spectra in both polarizations, relative error $\leq 7\times10^{-16}$]

2.2 Optimal coefficients and the focal matrix

The field at the focus in a (possibly complex) polarization $\hat{\mathbf{q}}$ is $\hat{\mathbf{q}}^*\cdot\mathbf{E}(0) = \sum_p \int \mathrm{d}^2\mathbf{k}_\parallel\, c_p\,(\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p)$. We consider two normalizations,

$$\text{power:}\quad \frac{2\pi^2}{Z_0}\sum_p\int \mathrm{d}^2\mathbf{k}_\parallel\,\frac{k_z}{k}|c_p|^2 = P, \qquad\qquad \text{amplitude:}\quad \frac{2\pi^2}{Z_0}\sum_p\int \mathrm{d}^2\mathbf{k}_\parallel\,|c_p|^2 \equiv P_{\mathrm{amp}}, \tag{7}$$

i.e. weights $g = k_z/k$ and $g = 1$ in (4). The optimal coefficients are $c_p \propto g^{-1}(\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p)^*$, i.e. $c_p^{(\mathrm{opt})} \propto (k/k_z)(\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p)^*$ for power normalization, which is the tutorial's $\alpha_p^{(\mathrm{opt})} \propto \sqrt{k/k_z}(\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p)^*$ written for $\alpha_p = \sqrt{k_z/k}\,c_p$. The maximum is

$$\begin{aligned} \max\frac{\varepsilon_0|\hat{\mathbf{q}}^*\cdot\mathbf{E}(0)|^2}{P} &= \frac{\varepsilon_0 Z_0}{2\pi^2}\sum_p\int \mathrm{d}^2\mathbf{k}_\parallel\,\frac{k}{k_z}|\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p|^2 = \frac{1}{2\pi^2 c}\,\hat{\mathbf{q}}^\dagger\mathsf{M}\,\hat{\mathbf{q}}, \\ \mathsf{M} &= \int \mathrm{d}^2\mathbf{k}_\parallel\,\frac{k}{k_z}(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}}), \end{aligned} \tag{8}$$

using the completeness relation $\sum_p \hat{\mathbf{e}}_p\hat{\mathbf{e}}_p^\dagger = \mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}}$. For amplitude normalization the factor k/k_z is absent.

Solid-angle form. Since $|\mathbf{k}_\parallel| = k\sin\theta'$ and $\mathrm{d}|\mathbf{k}_\parallel| = k\cos\theta'\mathrm{d}\theta'$,

$$\mathrm{d}^2\mathbf{k}_\parallel = k^2\cos\theta'\;\mathrm{d}\Omega, \qquad \text{so} \qquad \mathrm{d}^2\mathbf{k}_\parallel\,\frac{k}{k_z} = k^2\;\mathrm{d}\Omega. \tag{9}$$

Power normalization therefore weights all directions *uniformly in solid angle*. This observation organizes everything that follows. In particular, the power-optimal field for target $\hat{\mathbf{q}}$ is

$$\mathbf{E}^{(\mathrm{opt})}(\mathbf{x}) \propto k^2\int_{\mathrm{cap}} \mathrm{d}\Omega\,(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{q}}\,e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}}, \tag{10}$$

a Herglotz wave with uniform density over the cap of allowed directions. The amplitude-optimal field has an extra factor $\cos\theta'$ in the integrand.

2.3 Focal-point intensity maxima

The azimuthal integrals of the off-diagonal entries of M ($\propto \cos\varphi' \sin\varphi'$, $\cos\varphi'$, $\sin\varphi'$) vanish, so M is diagonal with $\mathsf{M}_{xx} = \mathsf{M}_{yy}$, and

$$\begin{aligned}\mathsf{M}_{xx} &= 2\pi k^2 \int_0^\theta \sin\theta' \Big(1 - \tfrac{1}{2}\sin^2\theta'\Big)\mathrm{d}\theta' = \pi k^2 \Big[\tfrac{4}{3} - \cos\theta - \tfrac{1}{3}\cos^3\theta\Big],\\ \mathsf{M}_{zz} &= 2\pi k^2 \int_0^\theta \sin^3\theta' \mathrm{d}\theta' = 2\pi k^2 \Big[\tfrac{2}{3} - \cos\theta + \tfrac{1}{3}\cos^3\theta\Big].\end{aligned} \tag{11}$$

Since $\hat{\mathbf{q}}^\dagger \mathsf{M} \hat{\mathbf{q}} \le \lambda_{\max}(\mathsf{M})$ for any unit $\hat{\mathbf{q}}$, including complex (e.g. circular) polarizations, the global optimum is in-plane polarization, because

$$\mathsf{M}_{xx} - \mathsf{M}_{zz} = \pi k^2 \cos\theta \sin^2\theta \ \ge 0. \tag{12}$$

Collecting results in units of $k^2/(2\pi c)$, with $P \to P_{\mathrm{amp}}$ for the amplitude-normalized entries:

$$\frac{2\pi c}{k^2}\frac{\varepsilon_0 |\mathbf{E}(0)|^2}{P} \le \begin{cases} \frac{4}{3} - \cos\theta - \frac{1}{3}\cos^3\theta & \text{power, in-plane } \hat{\mathbf{q}} \text{ (global max.)} \\ 2\big(\frac{2}{3} - \cos\theta + \frac{1}{3}\cos^3\theta\big) & \text{power, } \hat{\mathbf{q}} = \hat{\mathbf{z}} \\ \sin^2\theta\,\big(1 - \frac{1}{4}\sin^2\theta\big) & \text{amplitude, in-plane} \\ \frac{1}{2}\sin^4\theta & \text{amplitude, } \hat{\mathbf{q}} = \hat{\mathbf{z}}, \end{cases} \tag{13}$$

[numerical check: `focal_intensity_bounds.py`: the 3×3 matrix built with *explicit* TE/TM vectors reproduces all four entries to 3×10^{-13}, off-diagonals $< 10^{-16}$; an independent Monte-Carlo discretization with random polarization bases agrees to its sampling error]

Limits. Paraxially all in-plane entries tend to θ^2 and the z entries vanish as $\theta^4/2$. At 2π solid angle the power-normalized maximum is $4/3$ for *every* polarization, giving $U(\mathbf{x}_0)/P \le k^2/(6\pi c)$. The amplitude-normalized values are $3/4$ (in-plane) and $1/2$ (z), so the polarization-independence at 2π holds for power normalization only.

Amplitude normalization versus physical power. The amplitude-normalized entries of (13) are intensities per unit P_{amp}, not per unit physical power. The ratio $\frac{4/3}{3/4} = 16/9 \approx 1.78$ between the two maxima therefore compares different denominators. The amplitude-optimal beam, $c_p = (\hat{\mathbf{x}}\cdot\hat{\mathbf{e}}_p)^*$, measured against its physical power (6), delivers

$$\frac{2\pi c}{k^2}\frac{\varepsilon_0 |\mathbf{E}(0)|^2}{P} = \frac{\sin^4\theta\,(1 - \frac{1}{4}\sin^2\theta)^2}{2(\frac{4}{15} - \frac{1}{6}\cos^3\theta - \frac{1}{10}\cos^5\theta)} \xrightarrow{\theta\to\pi/2} \frac{135}{128} \approx 1.055, \tag{14}$$

where the denominator follows from

$$\int \mathrm{d}^2\mathbf{k}_\parallel \frac{k_z}{k}\Big(1 - \frac{k_x^2}{k^2}\Big) = 2\pi k^2 \int_{\cos\theta}^1 u^2 \Big(\frac{1}{2} + \frac{u^2}{2}\Big)\mathrm{d}u.$$

At equal physical power, then, the power-optimal beam exceeds the amplitude-optimal one by $(4/3)/(135/128) = 1.264$ at 2π solid angle. [numerical check: `focal_intensity_bounds.py`: explicit synthesis of the amplitude-optimal beam, rel. error $< 10^{-14}$]

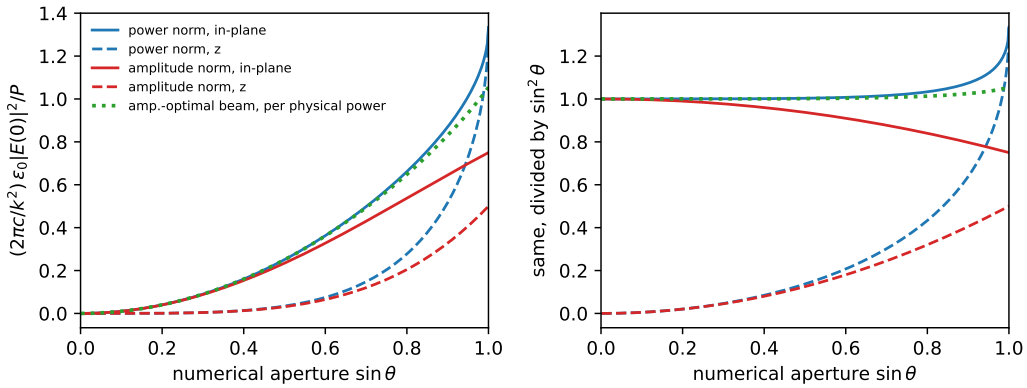


Figure 1: Maximum focal intensity versus numerical aperture, (13). The dotted curve is the amplitude-optimal beam measured per unit physical power, (14).

Reference beams and comparison with Dias & García de Abajo. For a family of reference beams $c_p \propto (k_z/k)^n(\hat{\mathbf{x}}\cdot\hat{\mathbf{e}}_p)^*$, the same algebra gives, at 2π solid angle and per unit physical power,

$$\frac{2\pi c}{k^2}\frac{\varepsilon_0|\mathbf{E}(0)|^2}{P} = \frac{\left[\frac{1}{n+2}+\frac{1}{n+4}\right]^2}{\frac{1}{2n+3}+\frac{1}{2n+5}} = \begin{cases} 4/3 & n=-1 \text{ (power-optimal)} \\ 1.2094 & n=-\frac{1}{2} \text{ (aplanatic lens, uniform pupil)} \\ 135/128 & n=0 \text{ (amplitude-optimal)} \\ 0.9292 & n=\frac{1}{2}. \end{cases} \tag{15}$$

The last case is the "diffraction-limited Gaussian" reference of Dias and García de Abajo (their $\beta \propto \hat{\mathbf{n}}\cdot\hat{\mathbf{e}}\sqrt{k_z/k}$). The optimum exceeds it by 1.435, the factor ≈ 1.43 quoted in the tutorial. Against an aplanatic lens with uniformly filled pupil, $c \propto (k/k_z)^{1/2}$, the advantage is only 1.10. Their focal-intensity result, Eq. (4) of that paper, $\max|\mathbf{E}(0)|^2/P_{\rm inc} = 4\omega^2/3c^3$ for 4π in Gaussian units with $\mathbf{E}_{\rm phys} = 2\,\mathrm{Re}[\mathbf{E}e^{-i\omega t}]$, agrees exactly with $(2\pi c/k^2)\varepsilon_0|\mathbf{E}|^2/P = 8/3$ derived here: converting to Gaussian units gives $|\mathbf{E}|^2/P = 16\omega^2/3c^3$, and halving the amplitude convention divides this by 4. The time-averaged energy density in their convention is $(|\mathbf{E}|^2+|\mathbf{H}|^2)/4\pi$, not $/8\pi$, which accounts for the factor of 2 between their energy-density bound and that of Bassett (and (13)). [numerical check: `focal_intensity_bounds.py`: closed form vs. explicit synthesis for $n=-1,-\frac{1}{2},0,\frac{1}{2},1$, agreement to 6 digits]

3 The optimal fields across the focal plane

3.1 General numerical aperture

Take $\hat{\mathbf{q}} = \hat{\mathbf{x}}$, write $\hat{\mathbf{k}} = (\sin\theta'\cos\varphi', \sin\theta'\sin\varphi', \cos\theta')$ and let the field point be at cylindrical coordinates (ρ,ϕ,z), with $a = k\rho$. The Cartesian components of $(\mathbb{I}-\hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{x}}$ are

$$(1-\sin^2\theta'\cos^2\varphi',\ -\tfrac{1}{2}\sin^2\theta'\sin 2\varphi',\ -\sin\theta'\cos\theta'\cos\varphi').$$

Shifting $\varphi' \to \varphi'+\phi$ in (1) gives

$$\begin{aligned} \int \mathrm{d}\varphi'\, e^{iu\cos(\varphi'-\phi)}\{1,\ \cos^2\varphi'\} &= \{2\pi J_0(u),\ \pi[J_0(u)-\cos 2\phi\, J_2(u)]\}, \\ \int \mathrm{d}\varphi'\, e^{iu\cos(\varphi'-\phi)}\{\sin 2\varphi',\ \cos\varphi'\} &= \{-2\pi\sin 2\phi\, J_2(u),\ 2\pi i\cos\phi\, J_1(u)\}, \end{aligned} \tag{16}$$

with $u = a\sin\theta'$. Hence, with $w(\theta') = 1$ (power) or $|\cos\theta'|$ (amplitude normalization),

$$\begin{aligned} E_x &= \pi k^2 \int \mathrm{d}\theta'\, \sin\theta'\, w\, e^{ikz\cos\theta'} \Big[(2-\sin^2\theta')J_0(u) + \sin^2\theta'\cos 2\phi\, J_2(u)\Big], \\ E_y &= \pi k^2 \sin 2\phi \int \mathrm{d}\theta'\, \sin^3\theta'\, w\, e^{ikz\cos\theta'} J_2(u), \\ E_z &= -2\pi i k^2 \cos\phi \int \mathrm{d}\theta'\, \sin^2\theta'\cos\theta'\, w\, e^{ikz\cos\theta'} J_1(u), \end{aligned} \tag{17}$$

with $\theta' \in [0,\theta]$ for one-sided excitation, and in addition $\theta' \in [\pi-\theta,\pi]$ for two-sided excitation. The focal spot is *not* rotationally symmetric: E_x depends on $\cos 2\phi$, and $E_y \propto \sin 2\phi$ and $E_z \propto \cos\phi$ both vanish on the axes. In particular E_z is antisymmetric in x, with two lobes of opposite sign on either side of the focus along the polarization axis. [numerical check: `focal_plane_fields.py`: (17) vs. brute-force 2D quadrature with explicit polarization vectors, rel. error 2×10^{-14}]

3.2 Closed forms at 2π solid angle

For the power-optimal beam at $\theta = \pi/2$ and $z = 0$, substitute $\beta = \sin\theta'$, so that $\sin\theta'\mathrm{d}\theta' = \beta\,\mathrm{d}\beta/\sqrt{1-\beta^2}$. The integrals reduce to the cases (3), together with

$$\int_0^1 \frac{\beta^3 J_0(a\beta)}{\sqrt{1-\beta^2}}\mathrm{d}\beta = j_0(a) - \int_0^1 \beta\sqrt{1-\beta^2}J_0(a\beta)\mathrm{d}\beta = j_0(a) - \frac{j_1(a)}{a}.$$

With $j_2 = 3j_1/a - j_0$:

$$\begin{aligned} E_x &= 2\pi k^2\Big[\sin^2\phi\; j_0(a) + (3\cos^2\phi - 1)\frac{j_1(a)}{a}\Big], \\ E_y &= 2\pi k^2 \sin\phi\cos\phi\; j_2(a), \qquad E_z = -2\pi i k^2\cos\phi\,\frac{J_2(a)}{a}, \end{aligned} \tag{18}$$

with $E_x(0) = 4\pi k^2/3$. Comparing with the full-sphere result (21) below shows that the in-plane part of (18) is *exactly one half* of the 4π-optimal field in the focal plane. This follows from symmetry: under $z \to -z$ the lower cap maps to the upper one, the in-plane components of $(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{x}}$ are even in $\hat{k}_z$, and E_z is odd. Adding the mirror hemisphere therefore doubles $\mathbf{E}_\parallel$ and cancels E_z at $z = 0$. [numerical check: closed form vs. quadrature 1.6×10^{-14}; $|\mathbf{E}_\parallel^{2\pi} - \tfrac{1}{2}\mathbf{E}_\parallel^{4\pi}|/|\mathbf{E}| < 10^{-16}$]

Along the polarization axis ($\phi = 0$) and perpendicular to it ($\phi = \pi/2$):

$$|\mathbf{E}|^2_{\phi=0} = 4\pi^2k^4\Big[4\Big(\frac{j_1(a)}{a}\Big)^2 + \Big(\frac{J_2(a)}{a}\Big)^2\Big], \qquad |\mathbf{E}|^2_{\phi=\pi/2} = 4\pi^2k^4\big[j_0(a) - j_1(a)/a\big]^2. \tag{19}$$

The first is the expression printed in the tutorial, which is therefore the cut along the polarization axis. The second has *exactly the same shape* as the 4π profile in the plane perpendicular to the polarization (Sec. 4). The 2π-optimal spot is elliptical: its width along the polarization axis exceeds the paraxial Airy width, while its width perpendicular to the polarization is already the 4π value (Table 1).

Amplitude normalization, any NA. The extra $\cos\theta'$ turns the integrals into elementary Bessel integrals ($s = \sin\theta$, Bessel arguments as):

$$E_x = \pi k^2\Big[(2s - s^3)\frac{J_1}{a} + 2s^2\frac{J_2}{a^2} + \cos 2\phi\, s^3\frac{J_3}{a}\Big], \qquad E_y = \pi k^2\sin 2\phi\, s^3\frac{J_3}{a}, \tag{20}$$

and at $s = 1$, $E_z = -2\pi i k^2\cos\phi\, j_2(a)/a$ [last case of (3)]. At $\phi = 0$ the recurrences reduce E_x to $\pi k^2 s^2[(1-\frac{s^2}{4})J_0 + J_2 + \frac{s^2}{4}J_4]$. [numerical check: `tests/test_sec3_1.py`: rel. error $< 5\times10^{-14}$ at $s = 0.3, 0.8, 1$]

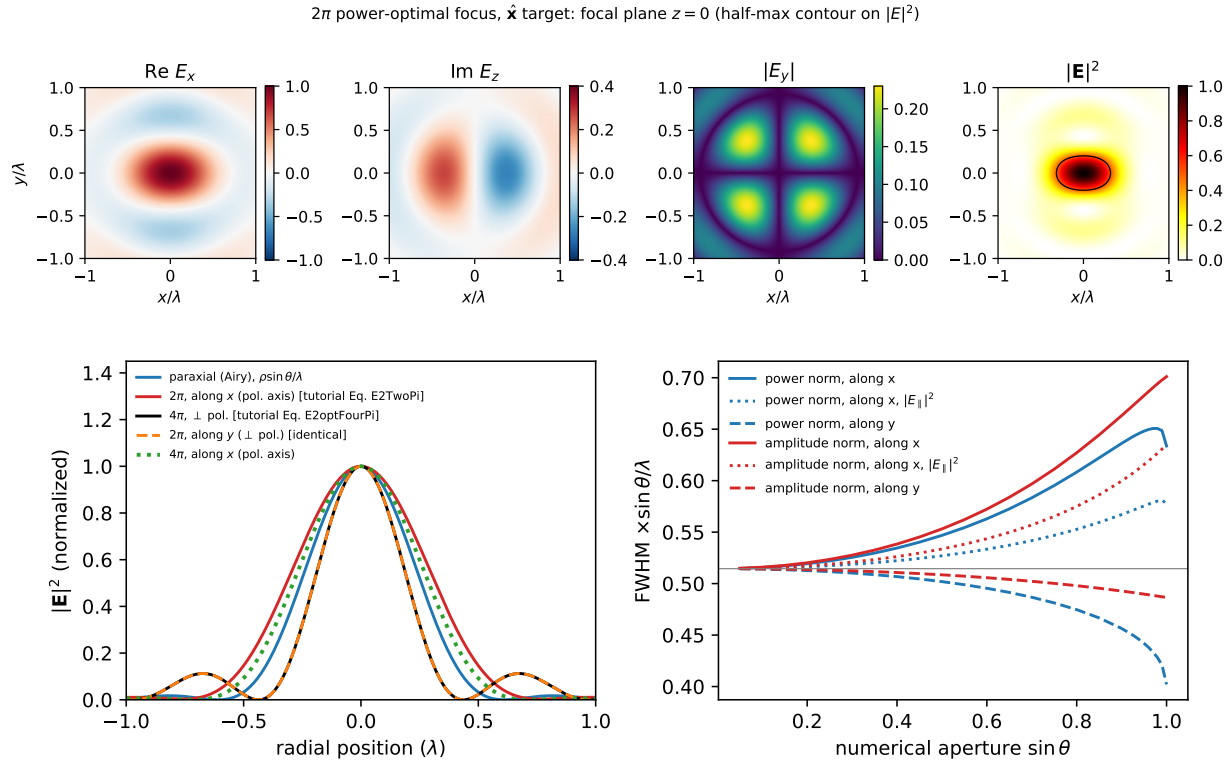


Figure 2: Top: focal-plane fields of the 2π power-optimal beam, (18). $E_z \propto \cos\phi$ is antisymmetric in x, and the half-maximum contour of $|\mathbf{E}|^2$ is an ellipse elongated along the polarization. Bottom left: intensity cuts. The 2π profile perpendicular to the polarization (dashed orange) coincides with the 4π profile (black). Bottom right: FWHM versus NA along the polarization axis (x, solid) and perpendicular to it (y, dashed); dotted curves show the in-plane intensity $|\mathbf{E}_\parallel|^2$ along x.

Paraxial limit. For $\theta \ll 1$ both normalizations give the Airy pattern,

$$E_x \simeq \int_{|\mathbf{k}_\parallel| \le k \sin\theta} \mathrm{d}^2\mathbf{k}_\parallel \, e^{i\mathbf{k}_\parallel \cdot \mathbf{x}_\parallel} = 2\pi k^2 \sin^2\theta \, \frac{J_1(a\sin\theta)}{a\sin\theta}, \qquad E_y, E_z \to 0.$$

Two features of Table 1 and Fig. 2 deserve emphasis:

1. In normalized units ($\rho\sin\theta/\lambda$), increasing the NA *broadens* the spot along the polarization axis, from 0.51 to 0.63, but *narrows* it perpendicular to the polarization, from 0.51 to 0.40. The width along x is non-monotonic very close to 2π (maximum 0.651 near $\sin\theta \approx 0.975$); this is a real feature of the optimal field, not a numerical artifact.

2. Relative to 2π, the 4π optimum is tighter in the focal plane only along the polarization axis, because the lower hemisphere cancels E_z (0.634 to 0.578). The large gain is along the optical axis (0.887 to 0.402). Perpendicular to the polarization the two profiles are identical.

Table 1: Full widths at half maximum of $|\mathbf{E}|^2$ and locations of the first minimum, in wavelengths (paraxial row in units of $\lambda/\sin\theta$). "First min." is a true zero only where $I/I_0 \approx 0$. The 4π beam is isotropic in the plane perpendicular to its polarization, so its y and z rows coincide.

excitation	direction	FWHM	first min.	I/I_0 there
paraxial (Airy)	any	0.5145	0.6098	0
2π, power norm.	x (polarization)	0.6337	0.7650	2.8×10^{-3}
	y	0.4021	0.4367	0
	z (optical axis)	0.8867	0.9898	1.6×10^{-2}
2π, amplitude norm.	x	0.7011	0.8734	1.4×10^{-3}
	y	0.4865	0.5600	0
	z	1.1862	–	–
4π (power norm.)	x (polarization)	0.5777	0.7151	0
	y, z	0.4021	0.4367	0

4 Full 4π solid angle (tutorial Sec. 3.1.3)

4.1 Herglotz form and the imaginary part of the Green's function

For the full sphere, (10) can be evaluated in closed form from

$$\int \mathrm{d}\Omega\, e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}} = 4\pi j_0(kr), \qquad \int \mathrm{d}\Omega\, \hat{k}_i\hat{k}_j e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}} = -k^{-2}\partial_i\partial_j[4\pi j_0(kr)].$$

Using $j_0' = -j_1$ and $j_1' = j_0 - 2j_1/x$ ($x = kr$):

$$k^2\int \mathrm{d}\Omega\,(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}} = 4\pi k^2\Big[\Big(j_0 - \frac{j_1}{x}\Big)\mathbb{I} + \Big(\frac{3j_1}{x} - j_0\Big)\hat{\mathbf{r}}\hat{\mathbf{r}}^{\mathrm{T}}\Big] = 16\pi^2 k\ \mathrm{Im}\,\mathsf{G}_0(\mathbf{x},0), \tag{21}$$

where $\mathsf{G}_0 = (\mathbb{I} + \nabla\nabla/k^2)e^{ikr}/(4\pi r)$ is the free-space dyadic Green's function. [numerical check: `fourpi_spherical_waves.py`: 8×10^{-15}] The 4π-optimal field is thus $\mathrm{Im}\,\mathsf{G}_0$ (the time-reversal field of Sec. 7.4), and the 2π-optimal field is "half of $\mathrm{Im}\,\mathsf{G}_0$", restricted to one hemisphere of directions. In the plane perpendicular to $\hat{\mathbf{q}}$ ($\hat{\mathbf{r}}\cdot\hat{\mathbf{q}} = 0$) the field is $4\pi k^2(j_0 - j_1/x)\hat{\mathbf{q}}$, isotropic in that plane. Along $\hat{\mathbf{q}}$ it is $8\pi k^2(j_1/x)\hat{\mathbf{q}}$.

4.2 Vector spherical waves

Define normalized vector spherical waves $\mathbf{M}_{\ell m} = z_\ell(kr)\mathbf{X}_{\ell m}$ and $\mathbf{N}_{\ell m} = k^{-1}\nabla\times\mathbf{M}_{\ell m}$, with $\mathbf{X}_{\ell m} = \mathbf{L}Y_{\ell m}/\sqrt{\ell(\ell+1)}$. Explicitly,

$$\mathbf{N}_{\ell m} = i\left[\sqrt{\ell(\ell+1)}\,\frac{z_\ell(x)}{x}Y_{\ell m}\hat{\mathbf{r}} + \frac{1}{\sqrt{\ell(\ell+1)}}\,\frac{1}{x}\,\frac{\mathrm{d}[xz_\ell(x)]}{\mathrm{d}x}\nabla_s Y_{\ell m}\right], \tag{22}$$

with $\nabla_s = \hat{\boldsymbol{\theta}}\partial_\theta + \hat{\boldsymbol{\phi}}\sin^{-1}\theta\,\partial_\phi$ and $z_\ell \in \{j_\ell, h_\ell^{(1)}, h_\ell^{(2)}\}$. [numerical check: finite-difference $\nabla\times\mathbf{M}$ vs. $k\mathbf{N}$: 2×10^{-10}] For regular waves at $r \to 0$: $\mathbf{M}_{\ell m} \propto j_\ell \to 0$; the radial part of $\mathbf{N}_{\ell m}$ is $\propto j_\ell(x)/x$, and the tangential part $\propto x^{-1}\mathrm{d}(xj_\ell)/\mathrm{d}x \to (\ell+1)x^{\ell-1}/(2\ell+1)!!$. Both vanish unless $\ell = 1$. At the origin only the three electric-dipole waves survive, and

$$\mathbf{N}_{1,0}(0) = i\,\tfrac{1}{\sqrt{2}}\,\tfrac{2}{3}\sqrt{\tfrac{3}{4\pi}}\,(\cos\theta\,\hat{\mathbf{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}) = i\,\tfrac{2}{3}\sqrt{\tfrac{3}{8\pi}}\,\hat{\mathbf{z}}, \tag{23}$$

and similarly for $m = \pm 1$ along the spherical basis vectors. [numerical check: scan of all regular VSWs with $\ell \le 6$: only $(1, m)$ nonzero; $|\mathbf{N}_{1m}(0)| = 0.2303294330$ exactly] In the equatorial plane, $\mathrm{d}(xj_1)/\mathrm{d}x = xj_0 - j_1$ gives $\mathbf{N}_{1,0} = -i\sqrt{3/(8\pi)}\,(j_0 - j_1/x)\,\hat{\boldsymbol{\theta}}$, which is the tutorial's equatorial expression up to the overall phase convention.

Incoming power and the 4π bound. The regular wave splits as $\mathbf{N}^{\mathrm{reg}} = \frac{1}{2}(\mathbf{N}^{(1)} + \mathbf{N}^{(2)})$. In the far field $h_\ell^{(2)}(x) \to i^{\ell+1}e^{-ix}/x$, $\mathbf{N}^{(2)}$ is transverse with $\int |\mathbf{N}^{(2)}|^2 r^2\,\mathrm{d}\Omega \to k^{-2}\int |\nabla_s Y|^2\,\mathrm{d}\Omega/[\ell(\ell+1)] = k^{-2}$, and $|\mathbf{H}| = |\mathbf{E}|/Z_0$. Hence the incoming part $\frac{1}{2}\mathbf{N}_{1,0}^{(2)}$ carries $P_{\mathrm{in}} = \frac{1}{2}\cdot\frac{1}{4}\cdot\frac{1}{Z_0k^2} = 1/(8Z_0k^2)$, and

$$\frac{U(\mathbf{x}_0)}{P_{\mathrm{in}}} = \frac{\frac{1}{4}\varepsilon_0\cdot\frac{4}{9}\cdot\frac{3}{8\pi}}{1/(8Z_0k^2)} = \frac{k^2}{3\pi c}. \tag{24}$$

[numerical check: inward Poynting flux at $R = 3, 30, 300$: 0.1250000000; $U/P_{\mathrm{in}} = 0.1061032954 = 1/(3\pi)$] The three channels $\mathbf{N}_{1,m}$ have mutually orthogonal values at the origin and orthogonal fluxes. For any target $\hat{\mathbf{q}}$, (4) therefore gives $\max|\hat{\mathbf{q}}^*\cdot\mathbf{E}(0)|^2/P_{\mathrm{in}} = \sum_m |\hat{\mathbf{q}}^*\cdot\mathbf{N}_{1m}(0)|^2/P_1 = |\mathbf{N}_{10}(0)|^2/P_1$, which is independent of $\hat{\mathbf{q}}$.

Doubling, in the plane-wave picture. For a Herglotz wave $\mathbf{E} = \int \mathrm{d}\Omega\,\mathbf{a}(\hat{\mathbf{k}})e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}}$, stationary phase gives $\mathbf{E} \sim (2\pi/ikr)[\mathbf{a}(\hat{\mathbf{r}})e^{ikr} - \mathbf{a}(-\hat{\mathbf{r}})e^{-ikr}]$ as $r\to\infty$, so $P_{\mathrm{in}} = (2\pi^2/Z_0k^2)\int|\mathbf{a}|^2\,\mathrm{d}\Omega$. For a one-sided beam this equals the z-directed power (6), by (9). With $\mathbf{a} = k^2(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{q}}$: $|\mathbf{E}(0)|^2 = (8\pi k^2/3)^2$ and $P_{\mathrm{in}} = (2\pi^2/Z_0)(8\pi/3)k^2$ for the full sphere, so $(2\pi c/k^2)\varepsilon_0|\mathbf{E}(0)|^2/P_{\mathrm{in}} = 8/3$. Each hemisphere alone carries half the power and half the field, giving the 2π value $4/3$. The full-sphere Herglotz wave is proportional to $\mathbf{N}_{1,0}^{\mathrm{reg}}$ (ratio constant to 5×10^{-12}), and its incoming power computed from the VSW coefficient agrees with the stationary-phase value, $16\pi^3/3$, to all printed digits.

5 Super-oscillatory spots (tutorial Sec. 3.1.4)

5.1 Minimum energy under point constraints

For $E(x) = \int_{-k_m}^{k_m} c(k)e^{ikx}\mathrm{d}k$ the field energy is $\int|E|^2\mathrm{d}x = 2\pi\int|c|^2\mathrm{d}k$. Minimizing $\int|c|^2$ subject to $E(x_i) = b_i$ is a minimum-norm problem, with solution $c(k) = \sum_i \lambda_i e^{-ikx_i}$ and

$$\min\int|c|^2\mathrm{d}k = \mathbf{b}^\dagger\mathsf{G}^{-1}\mathbf{b}, \qquad \mathsf{G}_{ij} = \int_{-k_m}^{k_m} e^{ik(x_i-x_j)}\mathrm{d}k = \frac{2\sin k_m(x_i-x_j)}{x_i-x_j}. \tag{25}$$

Two points. The single constraint $E(x+\delta) - E(x) = 1$ gives $\min\int|c|^2 = 1/\int|e^{ik\delta}-1|^2\mathrm{d}k = \left[4k_m - 4\sin(k_m\delta)/\delta\right]^{-1} \to 3/(2k_m^3\delta^2)$. This is the tutorial's $E \gtrsim 1/\delta^2$, with its constant.

Three points. Imposing $E(0) = 1$ and $E(\pm\delta) = 0$, the optimum is even in k and the constraints are equivalent to $\int c = 1$ and $\int c\,(1-\cos k\delta) = 1$. The Gram matrix of $\{1, 1-\cos k\delta\}$ has entries $2k_m$, $k_m^3\delta^2/3 + O(\delta^4)$ and $k_m^5\delta^4/10 + O(\delta^6)$, so

$$\min\int|c|^2\mathrm{d}k \;\to\; \frac{2k_m}{2k_m\cdot k_m^5\delta^4/10 - k_m^6\delta^4/9} = \frac{45}{2k_m^5\delta^4}, \qquad \frac{\text{constrained}}{\text{unconstrained}} \to \frac{45}{(k_m\delta)^4}. \tag{26}$$

The unconstrained optimum (c uniform, energy $1/2k_m$) is the sinc beam, whose first zero is at $k_m\delta = \pi$, where the penalty is exactly 1. Halving the spot ($k_m\delta = \pi/2$) costs a factor of 5.28. The asymptotic formula overestimates this, predicting 7.4. [numerical check: `superoscillation.py`: exact vs. asymptote ratio 0.9998 at $k_m\delta = 0.03$]

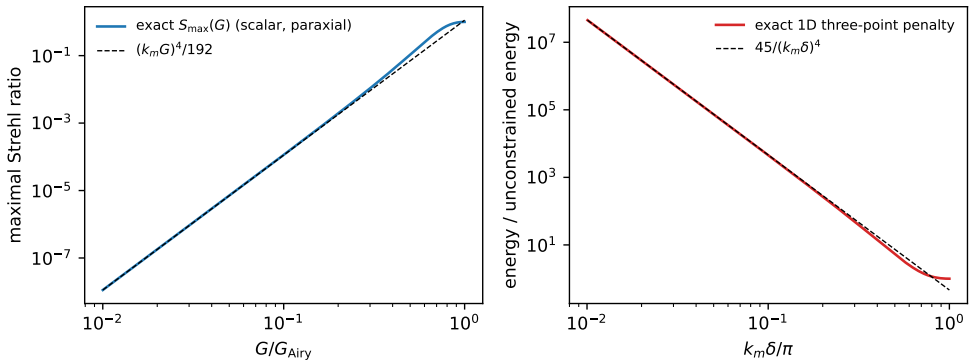


Figure 3: Left: exact maximal Strehl ratio for a zero ring (paraxial, scalar) and its $u^4/192$ asymptote. Right: exact 1D three-point energy penalty and its $45/(k_m\delta)^4$ asymptote.

5.2 Zero ring in two dimensions

In the paraxial scalar model with a circular pupil $|\mathbf{k}_\parallel| \le k_m$, a zero ring at radius G constrains every angular harmonic of the pupil function. The focal amplitude involves only the $m = 0$ harmonic, so the optimum sets the others to zero, leaving the single constraint $\int c_0(k) J_0(kG) k \, \mathrm{d}k = 0$. Projecting the uniform pupil orthogonally to $J_0(kG)$ with the inner product $2\pi \int_0^{k_m} \cdot \, k \, \mathrm{d}k$, and using $\int_0^{k_m} J_0(kG) k \, \mathrm{d}k = k_m J_1(u)/G$ and $\int_0^{k_m} J_0^2(kG) k \, \mathrm{d}k = \frac{k_m^2}{2}[J_0^2(u) + J_1^2(u)]$ with $u = k_m G$, the maximal Strehl ratio is

$$S_{\max}(u) = 1 - \frac{4J_1^2(u)}{u^2[J_0^2(u) + J_1^2(u)]} \xrightarrow{u\to 0} \frac{u^4}{192}, \tag{27}$$

with $S_{\max} = 1$ at the Airy zero $u = 3.8317$. The expansion follows from $[2J_1(u)/u]^2 = 1 - \frac{u^2}{4} + \frac{5u^4}{192}$ and $J_0^2 + J_1^2 = 1 - \frac{u^2}{4} + \frac{6u^4}{192}$. In terms of the spot size normalized to the Airy zero, $S_{\max} \approx 1.12\,(G/G_{\mathrm{Airy}})^4$. The exact penalties are $1/S_{\max} = 9.3$ for a factor-of-2 reduction and 8.7×10^3 for a factor of 10. The naive $2^4 = 16$ therefore overstates the cost of a factor-of-2 reduction by about 1.7×; $S_{\max}(0.5) \approx 0.1$ is also what the $S_{\max}$ curve of the tutorial's super-resolution figure shows. [numerical check: closed form vs. brute-force pupil discretization: $< 10^{-8}$]

6 Space-time focusing (tutorial Sec. 3.1.5)

Under amplitude normalization $\int\int |c(\mathbf{k}, \omega)|^2 = 1$, (4) makes the optimal c uniform on the support. For the 1D line (transverse wavenumber $|k| \le |\omega|/c$, $|\omega| \le \omega_m$):

$$\begin{aligned}\psi(x,t) &= \int_{-\omega_m}^{\omega_m} \mathrm{d}\omega \, \frac{2\sin(|\omega|x/c)}{x} e^{-i\omega t} = \frac{2}{x}\int_0^{\omega_m} \Big[\sin\omega\Big(\frac{x}{c} + t\Big) + \sin\omega\Big(\frac{x}{c} - t\Big)\Big]\mathrm{d}\omega \\ &= \frac{4c}{x}\Big[\frac{\sin^2(\omega_m(x+ct)/2c)}{x+ct} + \frac{\sin^2(\omega_m(x-ct)/2c)}{x-ct}\Big],\end{aligned} \tag{28}$$

which is the tutorial's result. At $t = 0$ it is $\propto \sin^2(\omega_m x/2c)/x^2$, with first zero at $2\pi c/\omega_m$, twice that of $\sin(\omega_m x/c)/x$. [numerical check: `spacetime_focusing.py`: vs. direct (ω, k) quadrature, 1.5×10^{-14}]

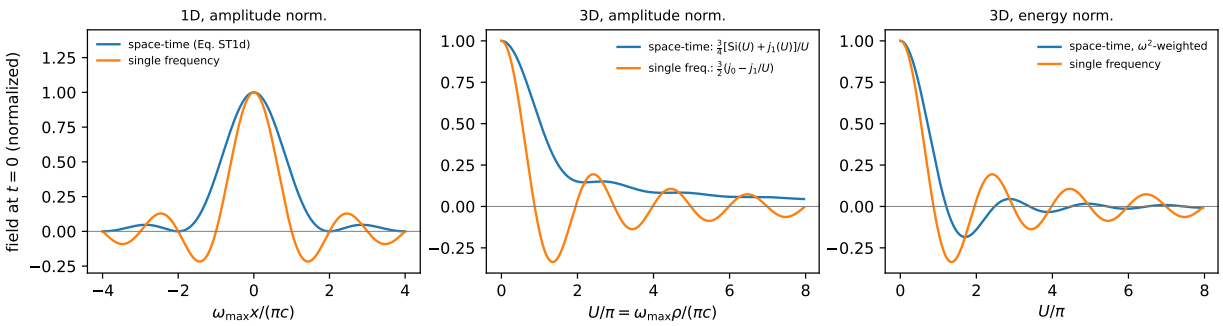


Figure 4: $t = 0$ profiles of space-time-optimal fields (blue) versus single-frequency optima at $\omega_{\max}$ (orange), for the 1D line and 3D, and in 3D for energy normalization.

Three dimensions. With uniform weight in ω, the $t = 0$ radial profile is $\int_0^{\omega_m}[j_0 - j_1/u]_{u=\omega\rho/c}\mathrm{d}\omega = (c/\rho)\int_0^U (j_0 - j_1/u)\mathrm{d}u$ with $U = \omega_m\rho/c$. Using $\int_0^U j_1(u)/u\,\mathrm{d}u = \frac{1}{2}[\mathrm{Si}(U) + \cos U/U - \sin U/U^2]$:

$$\psi_{3\mathrm{D}}(\rho, 0) \propto \frac{\mathrm{Si}(U) + j_1(U)}{U} = \frac{4}{3}\Big[1 - \frac{U^2}{15} + \ldots\Big] \quad (U \to 0), \qquad \to \frac{\pi}{2U} \quad (U \to \infty). \tag{29}$$

Because $\mathrm{Si}(U) > |j_1(U)|$ for all $U > 0$, this profile has *no zeros* and a non-oscillatory $1/\rho$ tail. Its intensity half width at half maximum is $1.80\times$ that of the single-frequency profile $j_0 - j_1/U$. The 1D features (faster side-lobe decay, first zero doubled) are therefore specific to the 1D line. [numerical check: closed form vs. quadrature 10^{-14}; minimum over $0 < U \le 200$ is $+5.9 \times 10^{-3}$]

Dependence on the normalization. Physical energy normalization changes the optimal weights. On the 2D line the per-frequency cost is $\int (k_z/k)|c|^2\mathrm{d}k_x$, so $c \propto k/k_z$ and the $t = 0$ profile becomes $\int_0^{\omega_m} \omega J_0(\omega x/c)\mathrm{d}\omega \propto J_1(U)/U$ (versus single-frequency $J_0(U)$; first-zero ratio 1.593). In 3D the incoming energy of a Herglotz pulse is $\propto \int \mathrm{d}\omega \int \mathrm{d}\Omega |\mathbf{a}|^2/\omega^2$, so $\mathbf{a} \propto \omega^2$ and the profile is $\propto [2\sin U - U\cos U - \mathrm{Si}(U)]/U^3$, which does have zeros (first at $U = 3.797$, versus 2.744 for a single frequency). The qualitative conclusion, that temporal localization costs spatial localization, survives. The factor of 2 and the side-lobe behavior do not.

7 Focusing in complex media (tutorial Sec. 3.2)

7.1 The bound and the optimal currents

Currents s on an aperture A produce $\psi(\mathbf{x}_0) = \Gamma_{0A}s$. For a target polarization ϕ, the objective $|\phi^\dagger\Gamma_{0A}s|^2$ with $\|s\| = 1$ is rank one, so by (4)

$$\max|\phi^\dagger\psi(\mathbf{x}_0)|^2 = \|\Gamma_{0A}^\dagger\phi\|^2 = \int_A |(\Gamma_{0A}^\dagger\phi)(\mathbf{x}')|^2\mathrm{d}\mathbf{x}', \qquad s_{\mathrm{opt}} \propto \Gamma_{0A}^\dagger\phi. \tag{30}$$

Nothing about the medium has been assumed. If the medium is reciprocal, $\Gamma(\mathbf{x}_0, \mathbf{x}')^{\mathrm{T}} = \Gamma(\mathbf{x}', \mathbf{x}_0)$, then $(\Gamma_{0A}^\dagger\phi)(\mathbf{x}') = [\Gamma(\mathbf{x}', \mathbf{x}_0)\phi^*]^*$, the conjugate of the field on A radiated by a dipole ϕ^* at $\mathbf{x}_0$. If the medium is not reciprocal, generalized reciprocity $\Gamma_\nu(\mathbf{x}, \mathbf{x}')^{\mathrm{T}} = \Gamma_{\nu^{\mathrm{T}}}(\mathbf{x}', \mathbf{x})$ gives the same recipe in the *transposed* medium.

Table 2: Achieved focal intensity divided by the bound, for currents prescribed by the conjugated dipole field computed in the actual medium ("recipe") or in the transposed medium (`complex_media_focusing.py`).

	open near-field patch		closed sphere, $R = 6\lambda$	
medium	recipe	transposed	recipe	transposed
lossless, reciprocal	1.00000000	1.00000000	1.00000000	1.00000000
lossy, reciprocal	1.00000000	1.00000000	1.00000000	1.00000000
TRI, non-reciprocal	0.216	1.00000000	0.084	1.00000000
gyrotropic	0.265	1.00000000	0.231	1.00000000

Far-field aperture and the LDOS. For a closed aperture in the far zone of a lossless medium, $|\mathbf{E}| = Z_0|\mathbf{H}|$ and $\int_A |\mathbf{E}_0|^2 \mathrm{d}A = 2Z_0 P_{\mathrm{rad}}$, where $P_{\mathrm{rad}} = \frac{\omega}{2}\,\mathrm{Im}(\mathbf{p}^* \cdot \mathbf{E}(\mathbf{x}_0))$ is the power radiated by the dipole, i.e. the LDOS. For apertures in the near field the bound is larger.

7.2 A point-dipole medium with controllable symmetries

To test these statements we use electric point scatterers $\mathbf{p}_j = \alpha_j \mathbf{E}_{\mathrm{loc}}(\mathbf{x}_j)$ with $\alpha_j^{-1} = \alpha_{0,j}^{-1} - i\frac{k^3}{6\pi\varepsilon_0}\mathbb{I}$, where the radiation-reaction term is $\varepsilon_0^{-1}k^2\,\mathrm{Im}\,\mathsf{G}_0(0)$. With $\mathbf{E} = \varepsilon_0^{-1}k^2\mathsf{G}_0\mathbf{p}$, the medium Green's function is

$$\Gamma(\mathbf{x},\mathbf{x}') = \frac{k^2}{\varepsilon_0}\Big[\mathsf{G}_0(\mathbf{x},\mathbf{x}') + \frac{k^2}{\varepsilon_0}\mathsf{G}_0(\mathbf{x},S)\,\mathsf{T}\,\mathsf{G}_0(S,\mathbf{x}')\Big], \qquad \mathsf{T} = \big(\,\mathrm{diag}\,\alpha_j^{-1} - \tfrac{k^2}{\varepsilon_0}\mathsf{G}_{SS}\big)^{-1}, \tag{31}$$

where G_{SS} couples distinct scatterers. All blocks of G_0 are symmetric and $\mathsf{G}_0(\mathbf{x},\mathbf{x}') = \mathsf{G}_0(\mathbf{x}',\mathbf{x})$, so $\mathsf{T}^{\mathrm{T}} = \mathsf{T}|_{\alpha\to\alpha^{\mathrm{T}}}$. The symmetries of the medium are then controlled entirely by α_0:

- *reciprocal* $\Leftrightarrow \alpha_0^{\mathrm{T}} = \alpha_0$. In general, $\Gamma_\alpha(\mathbf{x},\mathbf{x}')^{\mathrm{T}} = \Gamma_{\alpha^{\mathrm{T}}}(\mathbf{x}',\mathbf{x})$.
- *lossless* $\Leftrightarrow \alpha_0^\dagger = \alpha_0$. The power absorbed by dipole j is $\frac{\omega}{2}\mathbf{p}^\dagger[(\alpha_0^{-1})^\dagger - \alpha_0^{-1}]\mathbf{p}/(2i)$, the radiation-reaction term accounting exactly for the scattered power $\frac{\omega k^3}{12\pi\varepsilon_0}|\mathbf{p}|^2$.
- *time-reversal invariant* $\Leftrightarrow \alpha_0$ real. Under $(\mathbf{E},\mathbf{H},\mathbf{p},\mathbf{m}) \to (\mathbf{E}^*,-\mathbf{H}^*,\mathbf{p}^*,-\mathbf{m}^*)$ the constitutive relation $\mathbf{p} = \alpha_0\mathbf{E}_{\mathrm{loc}}$ maps to itself iff $\alpha_0^* = \alpha_0$. The radiation-reaction term is part of the outgoing boundary condition, not of the medium.

The four test media are lossless reciprocal (α_0 real symmetric), lossy reciprocal (complex symmetric), *TRI but non-reciprocal* (real, non-symmetric), and gyrotropic (Hermitian, non-symmetric: lossless, non-reciprocal, non-TRI).

7.3 Numerical results

Table 2 tests the bound and the recipes with random clusters of 16 scatterers in a λ^3 box, a random complex target polarization, and electric currents on either an open near-field patch or a closed sphere of radius 6λ. Γ is built directly from T, without invoking reciprocity. The iterative eigensolution of the full operator confirms the rank-one maximum in all cases (rel. difference $\le 2\times10^{-15}$).

For the LDOS connection, closed spheres about $\mathbf{x}_0$ in the lossless reciprocal medium give bound/$(2P_{\mathrm{rad}})$ = 94.8, 45.8, 1.012 and 1.00012 for R = 0.06, 0.3, 3 and 30λ. The sphere with $R = 0.3\lambda$ passes through the cluster, and any aperture is allowed.

7.4 Partial time reversal on a closed surface

In the representation $\psi = (\mathbf{E}, i\mathbf{H})$, $s = (i\mathbf{J}_e, -\mathbf{J}_m)$ of the tutorial, complex conjugation is time reversal: $\mathbf{E} \to \mathbf{E}^*$, $\mathbf{H} \to -\mathbf{H}^*$, $\mathbf{J}_e \to -\mathbf{J}_e^*$, $\mathbf{J}_m \to \mathbf{J}_m^*$ (for point dipoles, $\mathbf{p} \to \mathbf{p}^*$, $\mathbf{m} \to -\mathbf{m}^*$). The four-step construction of the tutorial proceeds as follows. The Love currents $\mathbf{J}_s = \hat{\mathbf{n}} \times \mathbf{H}$, $\mathbf{J}_{m,s} = -\hat{\mathbf{n}} \times \mathbf{E}$ on a closed surface S enclosing all sources and scatterers radiate $(\mathbf{E}, \mathbf{H})$ outside S and zero inside, in the actual medium, since the scatterers inside S see no field. Their negatives are the perfectly cancelling currents s_{pcc}. If the medium is TRI, conjugation gives $M\psi^* = s_0^* + s_{\mathrm{pcc}}^*$, so

$$\psi_{\mathrm{TR}} \equiv \Gamma s_{\mathrm{pcc}}^* = \psi^* - \Gamma s_0^* = \begin{cases} -2i \operatorname{Im} \psi & \mathbf{x} \text{ inside } S \ (s_0 \text{ real}), \\ -\Gamma s_0^* & \mathbf{x} \text{ outside } S. \end{cases} \tag{32}$$

The entrywise-$\operatorname{Im}\Gamma$ form of the tutorial therefore holds for field points inside S. Reciprocity was not used at any step.

The numerical test uses a real dipole inside a sphere S of radius λ, with one scatterer placed 0.05λ from S to test the claim that no far-field assumption is needed. Fields and Love currents are evaluated with a 180×360 Gauss–Legendre×trapezoid quadrature on S. Time-reversed currents are radiated in the medium, and the result is compared with $-2i \operatorname{Im} \mathbf{E}$ at 25 interior points:

medium	$\max \lvert\psi_{\mathrm{TR}} + 2i \operatorname{Im} \mathbf{E}\rvert / \max \lvert 2 \operatorname{Im} \mathbf{E}\rvert$ inside S	extinction-theorem residual
lossless, reciprocal (TRI)	7×10^{-8}	2×10^{-7}
TRI, non-reciprocal	1×10^{-6}	4×10^{-6}
lossy, reciprocal (not TRI)	0.45	2×10^{-7}
gyrotropic (not TRI)	1.24	1×10^{-6}

The residuals of the TRI cases fall with quadrature refinement, tracking the extinction-theorem residual (at 120×240 they are 5×10^{-6} and 6×10^{-5}). The failures are $O(1)$ and do not change. This supports two statements of the tutorial: that time reversal, not reciprocity, underlies the conjugate-field identity, and that the time-reversal recipe and the reciprocity recipe are distinct. In the TRI non-reciprocal medium the time-reversed field is exactly $-2i \operatorname{Im} \mathbf{E}$, yet the conjugate-field recipe reaches only 8–22% of the maximum focal intensity (Table 2).

7.5 Space-time version

With sources $s(\mathbf{x}, t)$ on A for $t \in [0, T]$, $\psi(\mathbf{x}_0, t_0) = \int_0^T \mathrm{d}t \int_A \Gamma_{\mathrm{ret}}(\mathbf{x}_0, t_0; \mathbf{x}, t) s(\mathbf{x}, t)$, which is again rank one. By time-domain reciprocity and time-translation invariance, $\Gamma_{\mathrm{ret}}(\mathbf{x}_0, t_0; \mathbf{x}, t)^{\mathrm{T}} = \Gamma_{\mathrm{ret}}(\mathbf{x}, t_0 - t; \mathbf{x}_0, 0)$. The optimal source is then $s(\mathbf{x}, t) \propto \psi_0(\mathbf{x}, t_0 - t)$: the field of a dipole fired at $(\mathbf{x}_0, 0)$, time-reversed. The bound is

$$\max \lvert \phi^{\mathrm{T}} \psi(\mathbf{x}_0, t_0) \rvert^2 = \int_{t_0 - T}^{t_0} \mathrm{d}\tau \int_A \lvert \psi_0(\mathbf{x}, \tau) \rvert^2 \mathrm{d}\mathbf{x}, \tag{33}$$

which reduces to the window $[0, T]$ written in the tutorial when the focal time is the end of the source window, $t_0 = T$.

8 Index of numerical checks

result	script (repository)	agreement
plane-wave power formula	`power_normalization_check.py`	10^{-16}
focal-intensity maxima, optimal polarization	`focal_intensity_bounds.py`	3×10^{-13}
amplitude-optimal beam per physical power	`focal_intensity_bounds.py`	10^{-14}
focal-plane fields, closed forms, $\frac{1}{2}$ of 4π	`focal_plane_fields.py`	2×10^{-14}
widths (Table 1)	`focal_plane_fields.py`	10^{-4} (root finding)
VSW amplitudes, P_{in}, $k^2/3\pi c$, $\operatorname{Im} \mathsf{G}_0$	`fourpi_spherical_waves.py`	10^{-10}–10^{-15}
super-oscillation penalties, $S_{\max}$	`superoscillation.py`	10^{-8}
space-time profiles	`spacetime_focusing.py`	10^{-14}
complex-media bound, recipes, TR identity	`complex_media_focusing.py`	10^{-15} / 10^{-7}

All scripts run in seconds on a laptop (Python 3 with NumPy, SciPy, Matplotlib); `pytest tests/` runs the regression versions.